Ashwin Chembu

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PROFILE

Full-stack product engineer connecting React and TypeScript interfaces with Python and Node.js services, authentication, and cloud data. Builds complete workflows with role permissions and layered validation.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL

Frontend: React, Next.js, Tailwind CSS, Capacitor, responsive interfaces

Backend / APIs: Node.js, Express, FastAPI, REST APIs, OAuth, WebSockets

Data / Cloud: MongoDB, Firestore, Firebase Authentication, AWS S3

Testing: Vitest, React Testing Library, Firebase emulators, Playwright

EXPERIENCE

Business Technology Solutions Associate | ZS Associates

South San Francisco, CA | Jun 2024-Present

- Built **React** and FastAPI RAG assistants with AWS document ingestion and Redis services to support enterprise knowledge-search workflows.
- Built Python and Flask endpoints connecting AWS S3 and Smartsheet APIs to automate document management and cloud data handoffs.
- Translated ambiguous stakeholder requirements into testable rules, API integrations, source-to-target mappings, and delivery documentation.
- Contributed to enterprise RAG evaluation and model workflows using FastAPI, Uvicorn, and pytest to validate applied-AI service behavior.

Founder and Lead Developer | Tallyrus

Jun 2023-Present

- Designed **React** interfaces, rubric workflows, and iterative feedback loops that let educators review essays with consistent scoring logic.
- Built **React/TypeScript** and Express flows with Google OAuth, MongoDB persistence, secure sessions, and protected application routes.
- Built Flask services with MongoDB persistence and S3 document storage to manage application data and support essay-feedback workflows.

Data Analyst / Web Development Lead | Tech4Good at UCSC

Mar 2023-Present

- Contributed 158 attributed commits to a collaborator-owned FigJam repository, implementing **TypeScript** chat, polling, and export workflows.
- Implemented 20+ attributed Node/Express REST routes with MongoDB persistence and logging for shared chat, text entry, and polling.
- Separated state across three MongoDB logical databases, using change streams and persisted responses to synchronize edits and votes.

PROJECTS

Sylk | Multi-Tenant Supply-Chain Platform

Private organization product | 2025-Present

- Built **React** and **TypeScript** shipment workflows with **Firebase** for organization onboarding, line items, fulfillment, files, and real-time chat.
- Enforced organization isolation with Firestore and Storage rules, separating retailer financial data from shared manufacturing workflows.
- Validated tenant permissions and product journeys with Vitest, Testing Library, **Firebase** emulators, and **Playwright** across supply-chain flows.
- Packaged the **React** product as a Capacitor Android app with native-safe routing, **Firebase** Authentication, and phone-specific navigation flows.

Lume | AI Meeting Intelligence

Dec 2025-Present

- Built **React** and Node/Express workflows with MongoDB and S3 to capture, store, search, and query meeting transcripts and project files.
- Designed a RAG engine with vector embeddings, intent routing, evidence extraction, and generator/critic refinement for citation-backed answers.
- Built responsive **React** interfaces and WebSocket interactions connecting meeting capture, document search, and reusable project context.

3D Commerce Experience | Interactive Storefront

Public UI prototype

- Built reusable **React** and **TypeScript** storefront components for product discovery, search, cart, checkout, account, and content pages.
- Built a Three.js product builder with GLB assets and Draco decoding for independent garment colors, material controls, and placement changes.
- Implemented reusable navigation, product cards, modals, and consent flows with Vite and Tailwind, then verified a successful production build.

EDUCATION

University of California, Davis | **B.S. Computer Science** | **GPA: 3.7** — Graduated Jun 2025

CERTIFICATIONS

AWS Certified Cloud Practitioner | Google Data Analytics Certificate

LINKS

Portfolio: https://ashchembu.com | LinkedIn: linkedin.com/in/ashwin-chembu-a5292b205 | GitHub: github.com/ashwinchembu